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(54) **PORTABLE WORKSTATION WITH INTEGRATED POWER SOURCE AND TILT-TOP CHASSIS**

(71) Applicant: **August Berres Corporation**, Lewes, DE (US)

(72) Inventors: **Robert Kroon**, South San Francisco, CA (US); **Robert Russell**, Kentwood, MI (US); **Sterling Pau**, Vancouver, WA (US)

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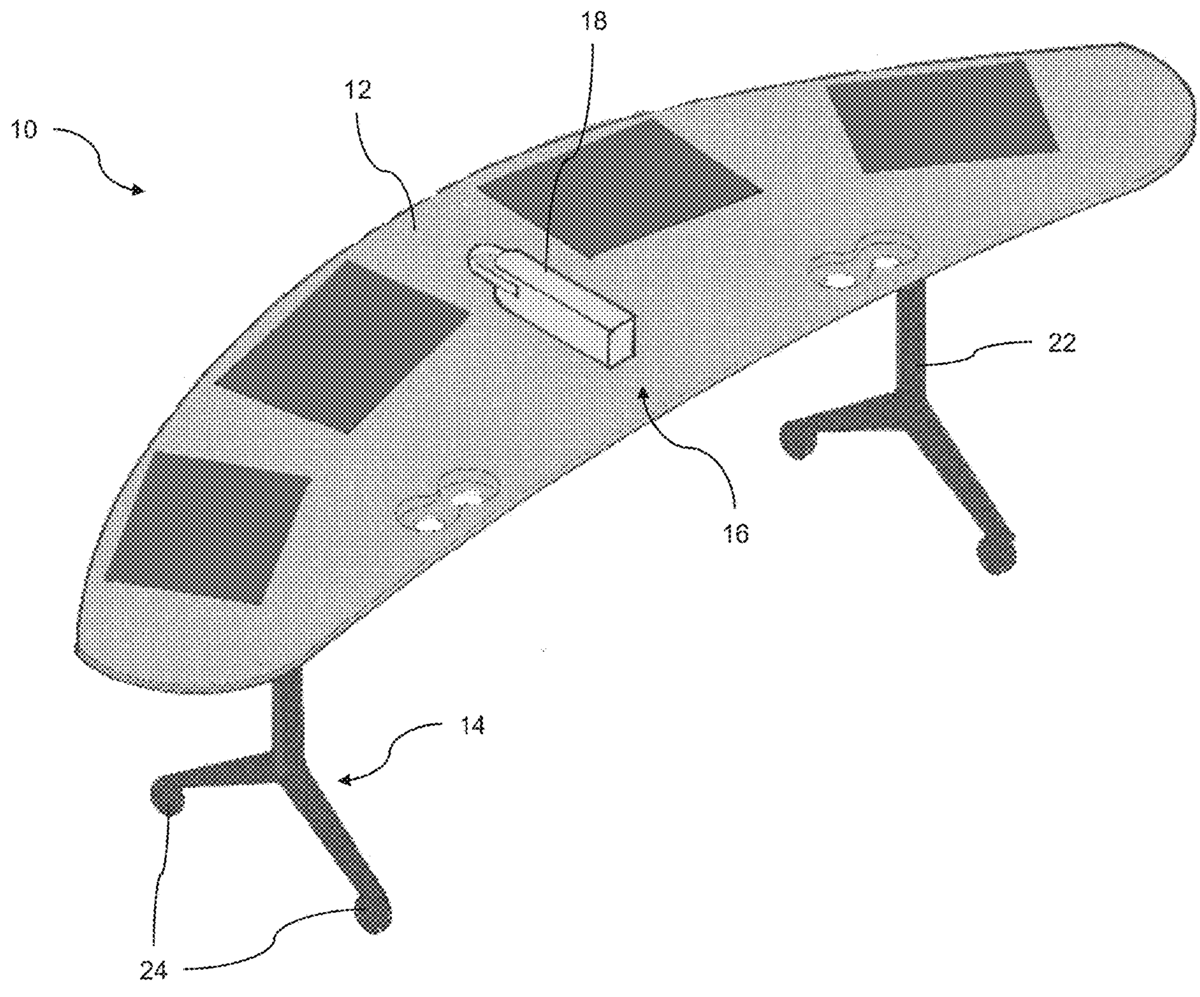
(57) **ABSTRACT**

A portable workstation includes a base that includes a plurality of wheels for moving the portable workstation within an environment. A panel is adjustably mounted at the base and is adjustable relative to the base between (i) a use position, where the panel is in a generally horizontal orientation, and (ii) a non-use position, where the panel is in a generally vertical orientation. A battery pack is accommodated at the portable workstation. The battery pack is configured to electrically power an electronic device at the portable workstation.



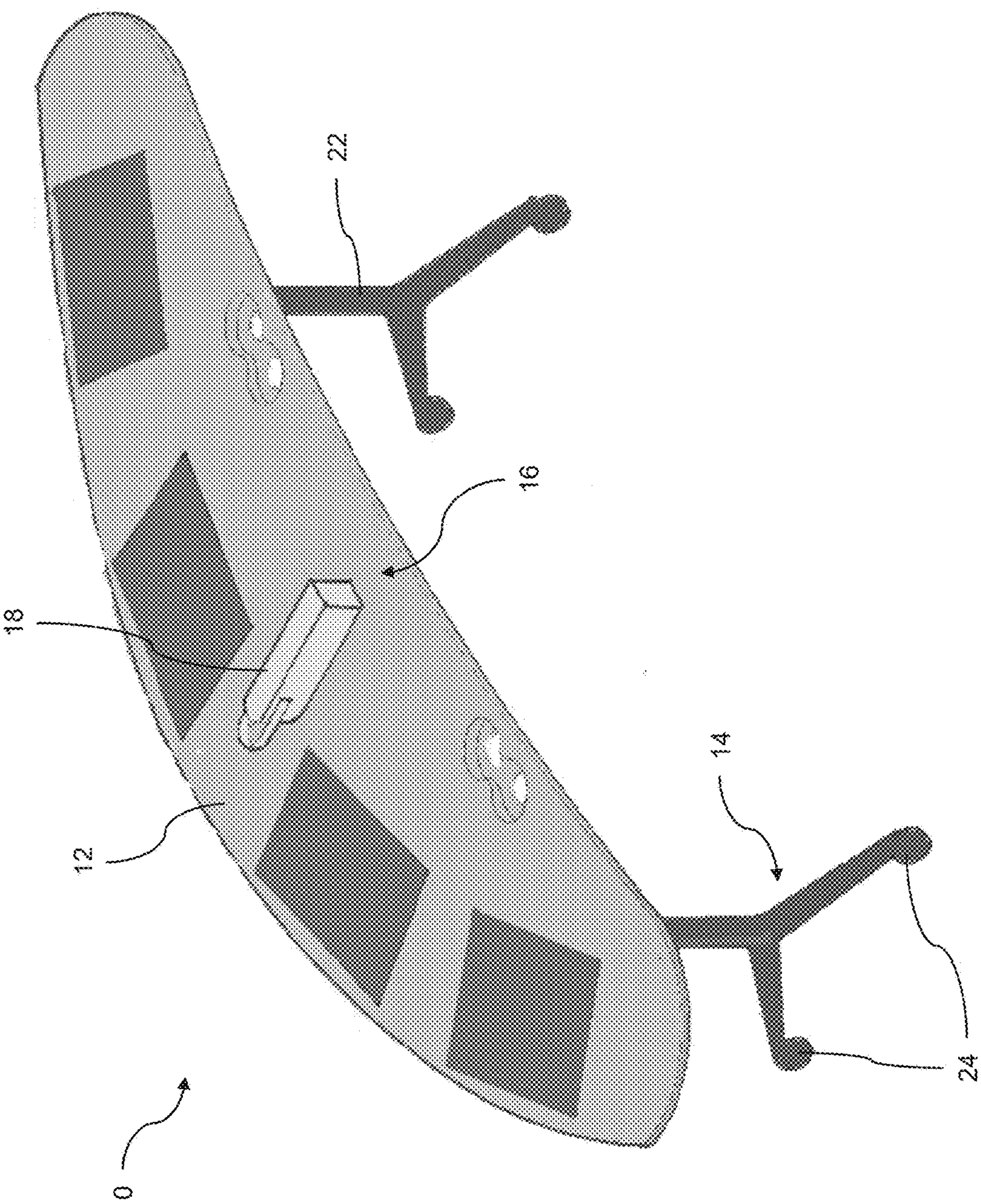


FIG. 1

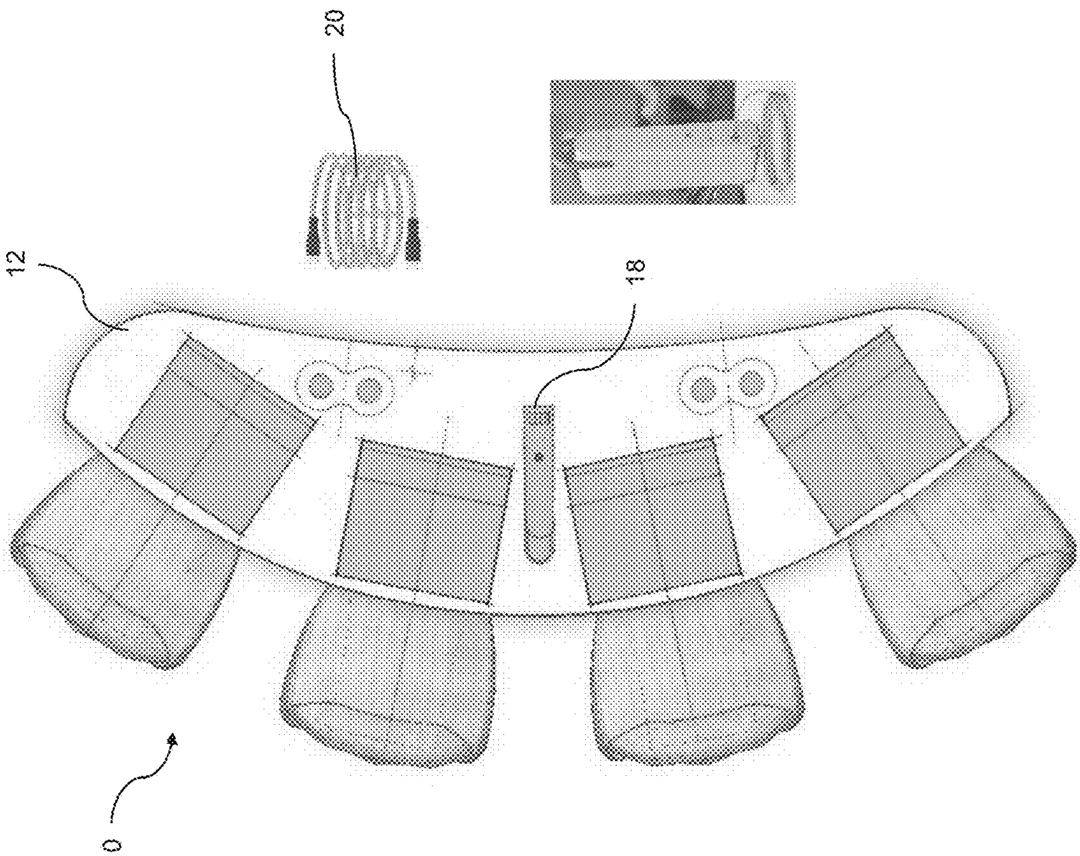


FIG. 2

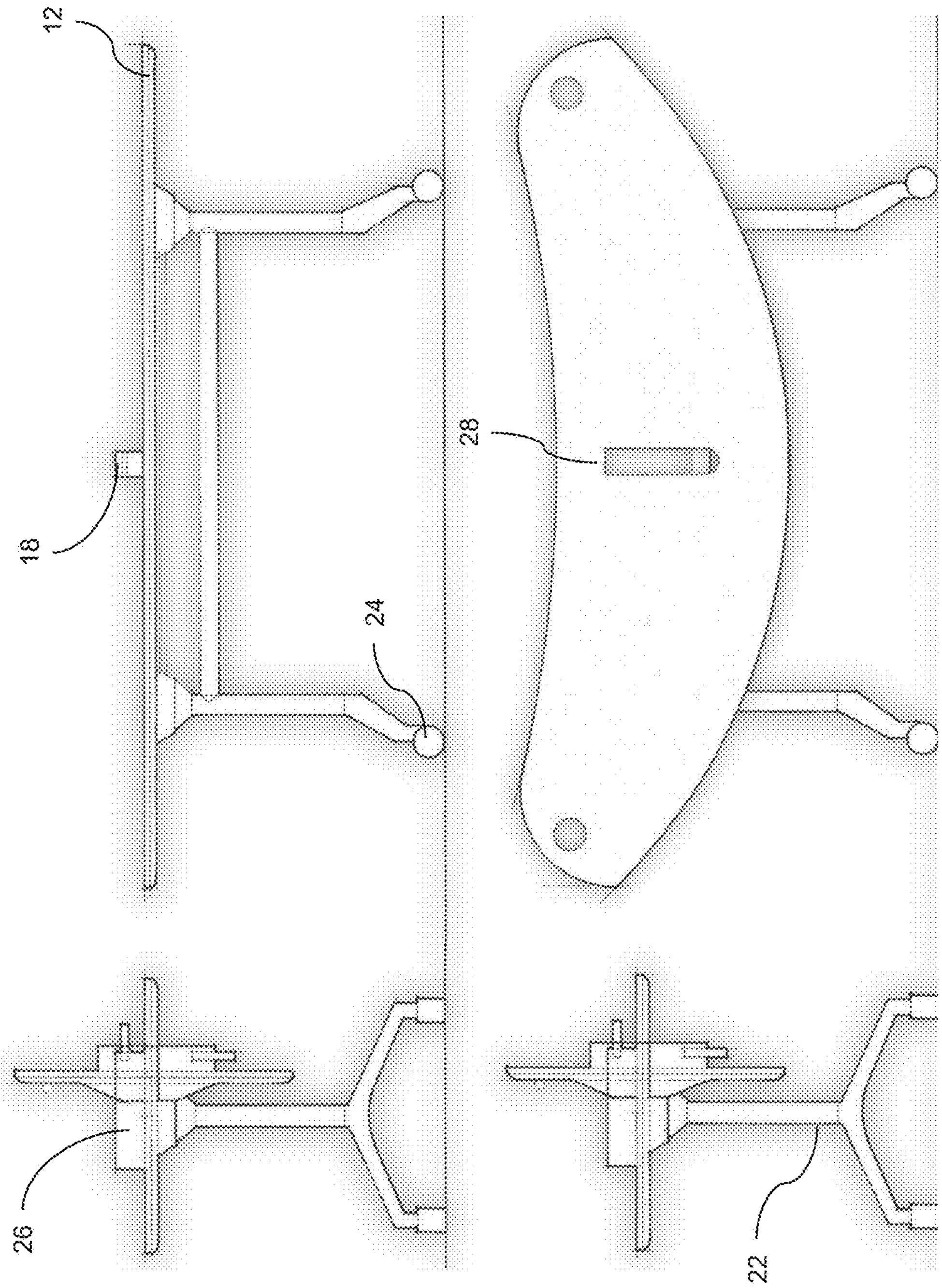


FIG. 3

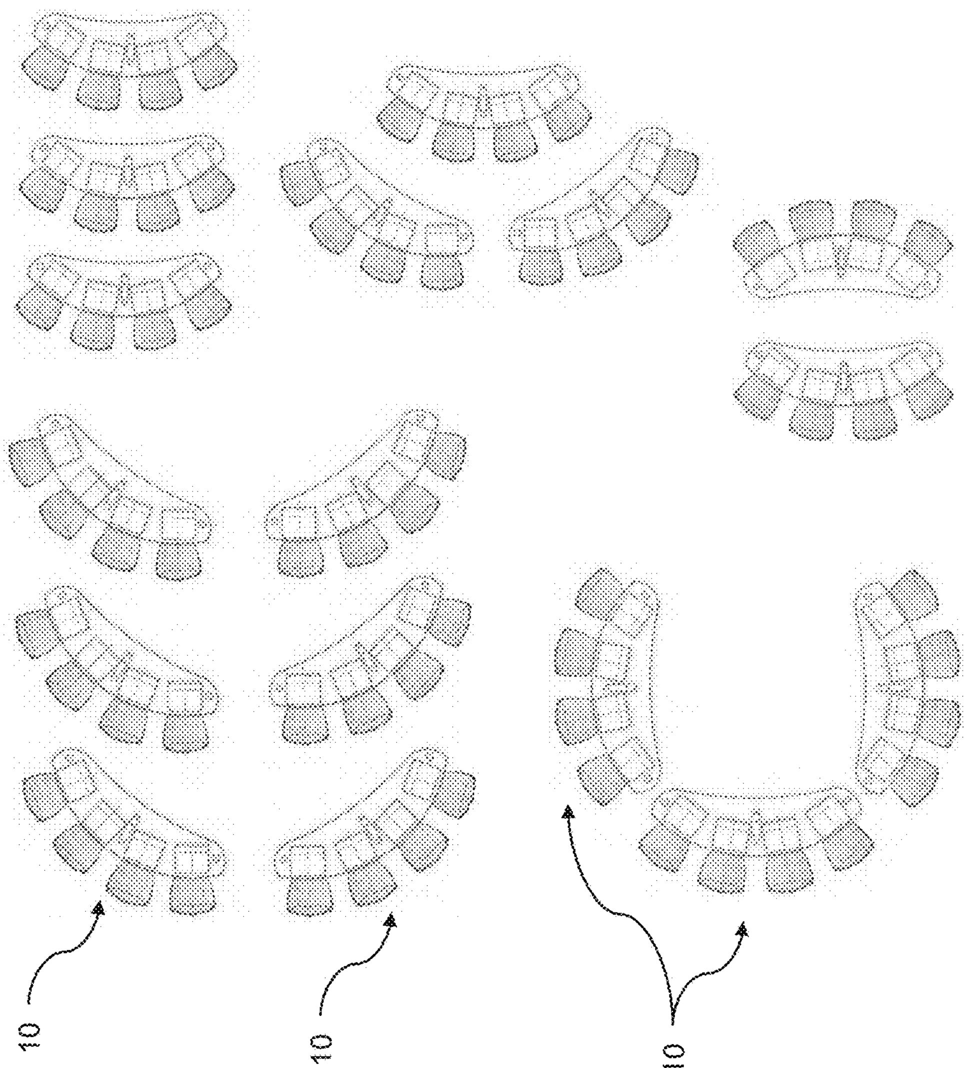


FIG. 4

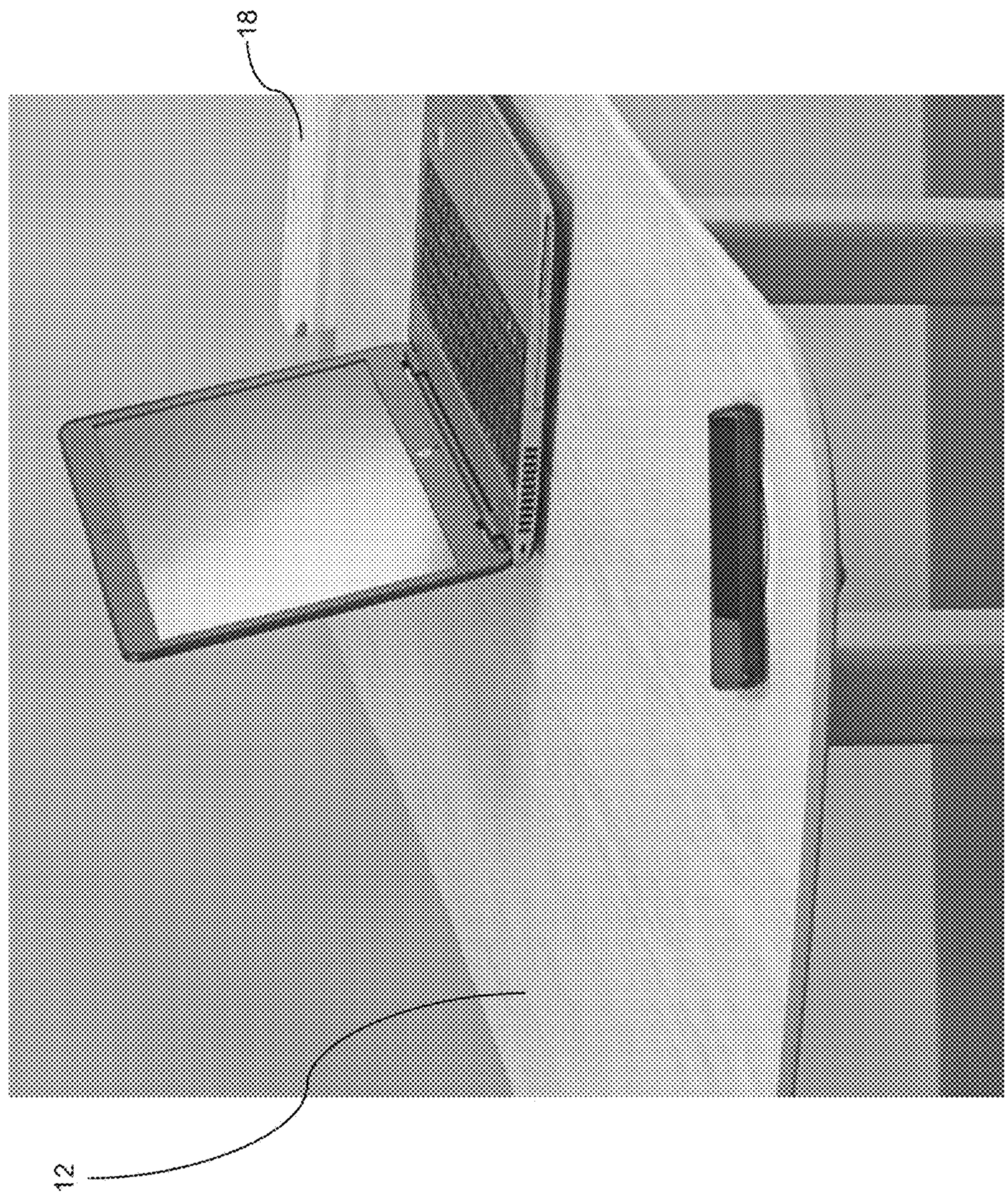


FIG. 5

32

FIG. 6

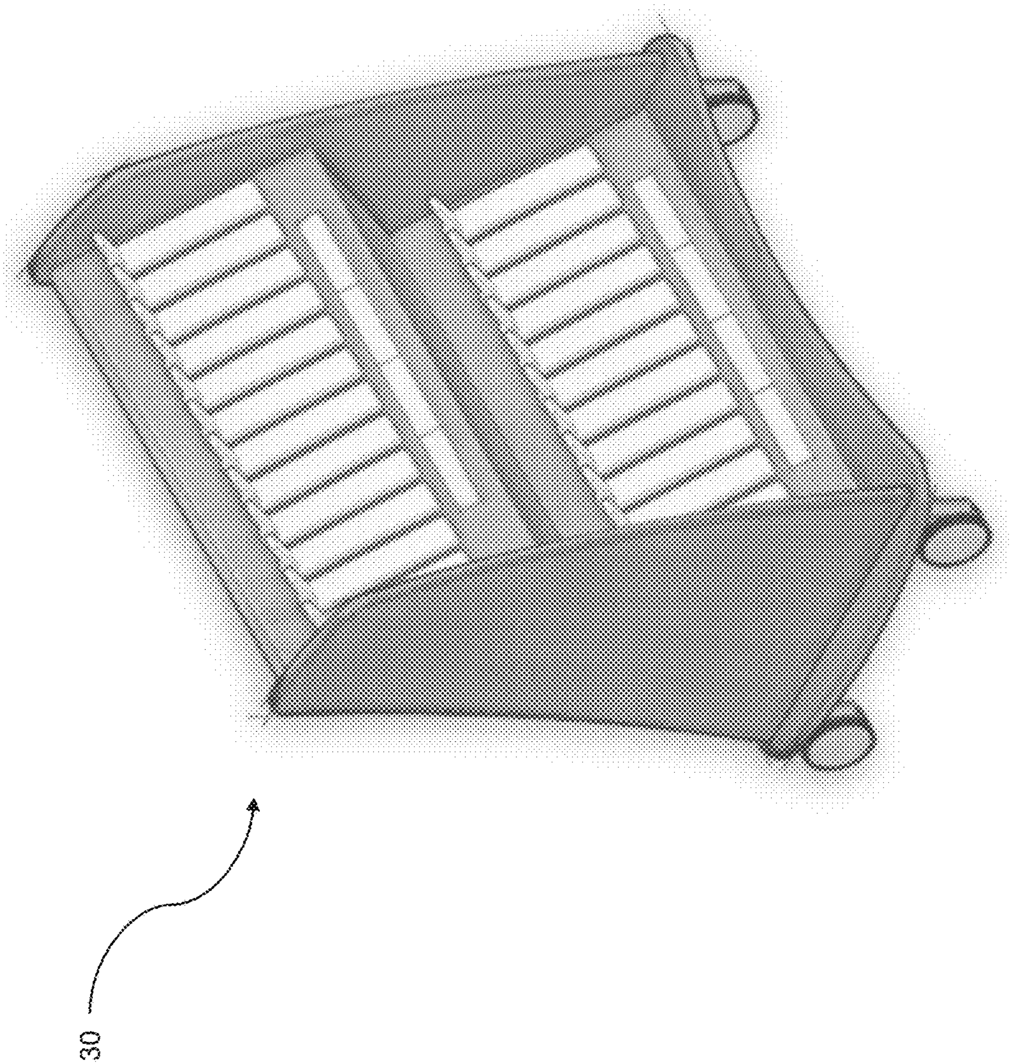


FIG. 7

PORTABLE WORKSTATION WITH INTEGRATED POWER SOURCE AND TILT-TOP CHASSIS

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application claims the filing benefits of U.S. provisional application Ser. No. 63/558,264, filed Feb. 27, 2024, which is hereby incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention relates generally to furniture and, more particularly, to portable workspace assemblies.

BACKGROUND OF THE INVENTION

[0003] Modular furniture is generally known in the art.

SUMMARY OF THE INVENTION

[0004] A portable workstation includes a base. The base includes a plurality of wheels for moving the portable workstation within an environment. A panel is adjustably mounted relative to the base between (i) a use position, where the panel is in a generally horizontal orientation, and (ii) a non-use position, where the panel is in a generally vertical orientation. A battery pack is accommodated at the portable workstation. The battery pack is configured to electrically power an electronic device at the workstation.

[0005] These and other objects, advantages, purposes and features of the present invention will become apparent upon review of the following specification in conjunction with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a perspective view of a portable workstation with integrated power source and tilt-top chassis;

[0007] FIG. **2** is a plan view of the portable workstation;

[0008] FIG. **3** shows the portable workstation adjusted between the use position and the non-use position;

[0009] FIG. **4** is a plan view of a plurality of portable workstations in an environment;

[0010] FIG. **5** shows the portable workstation with a single seating position and configured as a sit-stand desk;

[0011] FIG. **6** shows the portable workstation with integrated touchscreen display; and

[0012] FIG. **7** is a perspective view of a charging cart with docking stations configured to charge a plurality of battery packs associated with portable workstations.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0013] Laptop computers and other portable electronic devices dominate learning and the workplace. However, laptops and other portable electronic devices typically lack enough battery capacity for an entire day's use, presenting challenges and disruptions from power loss. While laptops may be recharged and/or powered via wired connection to a wall outlet, extension cords and other charging connectors extending between the wall outlet and a work station at which the laptop is being used can create a tripping hazard or other disruption to the environment. Thus, laptops and other portable electronic devices are typically recharged at the end of the school or work day, such as at home or at a storage and charging cart. That is, portable electronic devices (e.g., that are battery powered) are frequently utilized in classroom, work spaces, and other environments where access to an external power source (e.g., a wall outlet) may be limited.

[0014] Referring now to the drawings and the illustrated embodiments depicted therein, a workstation **10** includes a work surface **12**, such as a panel or tabletop, mounted at a base or chassis structure **14** and an integrated power system **16** configured to supply power to one or more electronic devices disposed at the work surface **12** (FIGS. **1** and **2**). For example, the power system **16** includes one or more battery packs or energy storage devices **18** operable to provide electric power to the one or more electronic devices at the work station **10**, such as via one or more outlets or ports (e.g., one or more electrical sockets and/or one or more USB ports) at the battery pack **18**. Because users are able to electrically power and charge their electronic devices directly from the battery pack **18** at the workstation **10**, users are able to maintain use of their electronic devices throughout work sessions (e.g., all day) without risk of disruption from power loss.

[0015] The work surface **12** may have any suitable shape, dimension and configuration. In the illustrated example, the work surface **12** comprises an arcuate shape so that users seated at respective seat positions along one side of the work surface **12** may be directed toward a common focal point (e.g., the front of a room). Optionally, the work surface **12** may have any suitable curved or circular or polygonal shape, such as a rectangle or oval. Further, the work surface **12** may be configured to accommodate any suitable number of seating positions along one or more sides of the work surface **12**, such as one or more seat positions, two or more seat positions, four or more seat positions (FIGS. **1** and **2**), and the like.

[0016] One or more battery packs **18** may mount at the workstation **10** to be directly accessible by users at the workstation **10**. That is, portable electronic devices may electrically connect directly to the battery pack **18** via outlets or sockets at the battery pack **18**. For example, the battery pack **18** may be received at a docking station disposed at or at least partially recessed into the upper surface of the work surface **12**. The battery pack **18** may be disposed at a central region of the work surface **12** or between respective seat positions of the workstation **10** to provide access to charge ports at the battery pack **18** for each of the users at the workstation **10**. In some examples, the workstation **10** may be equipped with a plurality of battery packs **18**, such as one for each seating position, one for each pair of seating positions, and the like.

[0017] Optionally, the one or more battery packs **18** may mount below the work surface **12**, such as at a lower surface of the work surface **12** or at a mounting portion of the base **14**. Integrated wiring or connections may electrically connect the battery pack **18** to one or more electrical outlets or sockets accessible at the work surface **12**. For example, electrical sockets may be disposed at the work surface **12** for receiving charging cords from portable electronic devices. Further, one or more charging cords **20**, such as self-coiling USB-C cables or other common charging cord types, may be disposed at the work surface **12** for supplying power from the battery pack **18** to portable electronic devices. Optionally, the charging cords **20** may be accommodated within housings recessed into the work surface **12** to keep the work surface **12** clear when the charging cords **20** are not being used. The charging cords **20** may be automatically retractable into the housings. Covers or lids may be disposed across the top of the housings to retain the charging cords therein. In the illustrated example, the workstation **10** is equipped with one charging cord **20** per seating position (i.e., four charging cords for four seating positions), and the workstation **10** may be equipped with any suitable number of charging cords **20**.

[0018] As shown in FIG. **3**, the work surface **12** is tiltable or pivotable relative to the base **14** between a use position, where the work surface **12** is in a generally horizontal orientation so that a user may rest belongings (e.g., portable electronic devices) on the work surface **12**, and a non-use position, where the work surface **12** is in a generally vertical orientation. The work surface **12** may be movable to the non-use position to assist in storage of the workstation **10** (e.g., the workstation may have a smaller width dimension in the non-use position as compared to the use position) and/or for easier movement of the workstation **10** during reconfiguration of the layout of a work environment.

[0019] For example, the base **14** may include a frame or structure **22** for supporting the work surface **12**, with casters or wheels **24** at lower ends of legs of the frame **22** to enable movement of the workstation **10** within the environment. Mounting portions or brackets **26** may be disposed at an upper portion of the frame **22** and configured to attach to the lower surface of the work surface **12**. The mounting brackets **26** may be pivotable relative to the frame **22** to enable pivoting of the work surface **12** between the use and non-use positions. Further, the work surface **12** may be tiltable to any position between the use position and the non-use position (e.g., any angle between horizontal and vertical, or 360 degrees of rotation), and lockable in the intermediate position.

[0020] As shown in FIG. **3**, the battery pack **18** mounts to the frame **22**, such as at a docking station disposed at the frame **22**, and is accessible through an aperture **28** formed through the work surface **12**. Thus, when the work surface **12** is pivoted, the battery pack **18** remains stationary at the frame **22** and the work surface **12** moves relative to the battery pack **18**, such that the battery pack **18** extends through an aperture in the work surface **12** when in the use position. Optionally, the battery pack **18** may be disposed at the frame **22** beneath the work surface **12**.

[0021] The battery pack **18** is removable from the workstation and replaceable or rechargeable. Because the battery pack **18** remains stationary during tilting of the work surface **12**, the battery pack **18** may not need to be locked or secured in place at the frame **22**, resulting in easier removal and replacement of the battery pack **18**. For example, a charging cart **30** (FIG. **7**) may include docking stations configured to receive and charge a plurality of battery packs **18**. The charging cart **30** may accommodate up to **24** battery packs **18** or more. Thus, battery packs **18** may be removed from each workstation **10** in an environment and charged overnight at a charging cart **30** to be replaced the next day.

[0022] For example, and as shown in FIG. **4**, a plurality of workstations **10** may be positioned within an environment (e.g., a classroom or an office space) in a variety of configurations. That is, workstations having different configurations (e.g., different sizes and therefore number of seating positions and/or different numbers or positions of battery packs) may be positioned within the environment and the workstations may be grouped or oriented in any suitable manner relative to one another and relative to other aspects of the environment. Pivoting the work surface **12** to the non-use position may improve the maneuverability of the workstation **10** during reconfigurations of the workstations **10** in the environment.

[0023] Further, the workstation **10** may be equipped with one or more integrated electronic components, such as a display screen (e.g., a touchscreen display screen) (FIG. **6**), one or more electrically operable motors for raising the work surface **12** relative to the base **14** (i.e., for sit-stand functionality), a light source, and the like. The one or more integrated electronic components may be electrically powered by the battery packs **18** disposed at the work station **10**, such as via electrical connection between the docking station and the electronic component. As shown in FIG. **6**, a touchscreen display **32** may be embedded with or integrally formed with or replace the work surface **12** such that the touchscreen display **32** is pivotable relative to the base **14**. That is, the touchscreen display **32** is pivotable relative to the base **14** between the horizontal position and the vertical position (and any adjustable intermediate position between the horizontal position and the vertical position) such that the user may use the touchscreen display in sitting or standing positions, or in a display mode. The work surface **12** may be locked or secured at a position between the horizontal and vertical position (e.g., FIG. **6**) so that a user may more easily view and/or interact with the touchscreen display **32**.

[0024] Thus, the work station **10** includes a tabletop or work surface **12** pivotably mounted to the chassis structure **22** to provide tilt-top functionality. Pivoting the work surface **12** to the non-use position aids in storage and portability (i.e., for reconfiguring layouts of work environments). Battery packs **18** are disposed at the work stations **10** for electrically powering one or more portable electronic devices and/or integrated electronic devices at the work station **10**. The battery packs **18** may remain stationary and/or secured relative to the base **14** as the work surface **12** is pivoted relative to the base **14** and battery pack **18**, with the battery pack **18** optionally extending through an aperture in the work surface **12** with the work surface in the use position. Moreover, the battery packs **18** are removable from the work station **10** for replacement and recharging. Further, the work stations **10** may be configured in any suitable manner, such as an instructors desk sized for one seating position and with a battery pack **18** and sit-stand functionality. Self-coiling USB-C cables may provide electrical connection between the battery pack **18** and the portable electronic device at the work station **10**. The battery packs **18** may provide up to 200 Wh of power or more each and are swappable without tools.

[0025] The workstation and integrated power system may include characteristics of the portable workstations, battery packs and power systems described in U.S. Publication Nos. US-2024-0402759; US-2024-0313560 and/or US-2022-0251832, which are hereby incorporated herein by reference in their entireties.

[0026] Various embodiments and optional designs and accessories are disclosed and shown herein. The scope of this disclosure is not limited to the various embodiments and optional designs and accessories. The dimensions and tolerances shown and discussed herein are intended to be exemplary of one or more particular designs of the assembly and are in no way intended to limit the scope of the particular embodiments disclosed herein.

[0027] Changes and modifications in the specifically described embodiments can be carried out without departing from the principles of the invention, which is intended to be limited only by the scope of the appended claims, as interpreted according to the principles of patent law including the doctrine of equivalents.

1. A portable workstation, the portable workstation comprising:

a base, wherein the base includes a plurality of wheels for moving the portable workstation within an environment;

a panel adjustably mounted at the base, wherein the panel is adjustable relative to the base between (i) a use position, where the panel is in a generally horizontal orientation, and (ii) a non-use position, where the panel is in a generally vertical orientation;

a battery pack disposed at the base, wherein the battery pack is configured to electrically power an electronic device at the portable workstation; and

wherein, with the panel in the use position, the battery pack at least partially extends through an aperture in the panel, and wherein, when the panel is adjusted between the use position and the non-use position, the panel moves relative to the base and relative to the battery pack.

2. The portable workstation of claim **1**, wherein the battery pack is removable from the portable workstation.

3. The portable workstation of claim **1**, wherein the battery pack is removably mounted at the base.

4. The portable workstation of claim **1**, wherein the battery pack is electrically connected to an electrical outlet at the panel for electrically connecting to the electronic device at the portable workstation.

5. The portable workstation of claim **1**, wherein, as the panel is adjusted between the use position and the non-use position, the battery pack is received through the aperture.

6. The portable workstation of claim **1**, wherein the battery pack is electrically connected to an electrical wire extending from the panel for electrically connecting to the electronic device at the portable workstation.

7. The portable workstation of claim **6**, wherein the electrical wire comprises a USB cable.

8. The portable workstation of claim **1**, wherein the electronic device comprises an electronic component of the portable workstation.

9. The portable workstation of claim **8**, wherein the electronic component comprises one selected from the group consisting of (i) a display screen at the panel, (ii) a motor for adjusting a height of the panel relative to the base and (iii) a light source of the portable workstation.

10. The portable workstation of claim **1**, wherein the electronic device comprises a portable electronic device.

11. The portable workstation of claim **1**, wherein the panel is adjustable relative to the base to one or more intermediate positions between the use position and the non-use position.

12. The portable workstation of claim **1**, wherein a plurality of battery packs are accommodated at the portable workstation, and wherein individual battery packs of the plurality of battery packs are spaced from one another at the portable workstation.

13. A portable workstation, the portable workstation comprising:

a base, wherein the base includes a plurality of wheels for moving the portable workstation within an environment;

a panel adjustably mounted at the base, wherein the panel is adjustable relative to the base between (i) a use position, where the panel is in a generally horizontal orientation, and (ii) a non-use position, where the panel is in a generally vertical orientation;

a battery pack disposed at the base, wherein the battery pack is configured to electrically power an electronic device at the portable workstation, and wherein the battery pack is removable from the portable workstation;

wherein the battery pack is electrically connected to an electrical outlet at the panel for electrically connecting to the electronic device at the portable workstation; and

wherein, with the panel in the use position, the battery pack at least partially extends through an aperture in the panel, and wherein, when the panel is adjusted between the use position and the non-use position, the panel moves relative to the base and relative to the battery pack.

14. The portable workstation of claim **13**, wherein, as the panel is adjusted between the use position and the non-use position, the battery pack is received through the aperture.

15. The portable workstation of claim **13**, wherein the electronic device comprises a portable electronic device.

16. The portable workstation of claim **13**, wherein the panel is adjustable relative to the base to one or more intermediate positions between the use position and the non-use position.

17. A portable workstation, the portable workstation comprising:

a base, wherein the base includes a plurality of wheels for moving the portable workstation within an environment;

a panel adjustably mounted at the base, wherein the panel is adjustable relative to the base between (i) a use position, where the panel is in a generally horizontal orientation, and (ii) a non-use position, where the panel is in a generally vertical orientation;

a plurality of battery packs disposed at the base, wherein individual battery packs of the plurality of battery packs are spaced from one another at the portable workstation, and wherein the individual battery packs of the plurality of battery packs are configured to electrically power one or more electronic devices at the portable workstation;

wherein the individual battery packs of the plurality of battery packs are removable from the portable workstation; and

wherein, with the panel in the use position, the individual battery packs of the plurality of battery packs at least partially extend through respective apertures in the panel, and wherein, when the panel is adjusted between the use position and the non-use position, the panel moves relative to the base and relative to the plurality of battery packs.

18. The portable workstation of claim **17**, wherein the individual battery packs of the plurality of battery packs are removably mounted at the base.

19. The portable workstation of claim **17**, wherein, as the panel is adjusted between the use position and the non-use position, the individual battery packs of the plurality of battery packs are received through the respective apertures.

20. The portable workstation of claim **17**, wherein the one or more electronic devices comprise one or more portable electronic devices.

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